

BINARY NUMBERS – SHORT REVISION NOTES

Remember Level (Recall Facts & Basic Knowledge)

1. Key Definitions

Term	Definition	Example
Binary Number System	A number system with base 2	1010_2
Base	The number of digits used in the system	Base 2
Digits Used	Only 0 and 1	0, 1
Decimal Number System	The number system with base 10 (Hindu-Arabic)	25_{10}
Place Value	The value of each position in a number	$2^3, 2^2, 2^1, 2^0$

2. Key Symbols

Symbol	Meaning	Example
$_2$ (subscript)	Subscript indicating base 2	1011_2
$_{10}$ (subscript)	Subscript indicating base 10	25_{10}
2^0	Two to the power of 0 = 1	$1 \times 2^0 = 1$
2^1	Two to the power of 1 = 2	$1 \times 2^1 = 2$
2^2	Two to the power of 2 = 4	$1 \times 2^2 = 4$
2^3	Two to the power of 3 = 8	$1 \times 2^3 = 8$
2^4	Two to the power of 4 = 16	$1 \times 2^4 = 16$
2^5	Two to the power of 5 = 32	$1 \times 2^5 = 32$
2^6	Two to the power of 6 = 64	$1 \times 2^6 = 64$
2^7	Two to the power of 7 = 128	$1 \times 2^7 = 128$
2^8	Two to the power of 8 = 256	$1 \times 2^8 = 256$

3. Powers of Two

Power	Value	Power	Value
2^0	1	2^6	64
2^1	2	2^7	128
2^2	4	2^8	256
2^3	8	2^9	512
2^4	16	2^{10}	1024
2^5	32	2^{11}	2048

Quick Memory Aid:

$$2^0 = 1$$

$$2^1 = 2$$

$$2^2 = 4$$

$$2^3 = 8$$

$$2^4 = 16$$

$$2^5 = 32$$

$$2^6 = 64$$

$$2^7 = 128$$

$$2^8 = 256$$

$$2^9 = 512$$

$$2^{10} = 1024$$

4. Place Values in Binary

Position	Place Value	Example: 1101_2
Rightmost (1st)	$2^0 = 1$	$1 \times 1 = 1$
2nd	$2^1 = 2$	$0 \times 2 = 0$
3rd	$2^2 = 4$	$1 \times 4 = 4$
4th	$2^3 = 8$	$1 \times 8 = 8$
5th	$2^4 = 16$	-
6th	$2^5 = 32$	-

Place Value Table:

Binary: 1 1 0 1
Position: 4th 3rd 2nd 1st
Place Value: 8 4 2 1
Value: $8 + 4 + 0 + 1 = 13$

5. Converting Decimal to Binary

Method: Repeated Division by 2

Step	Action	Example: $22_{10} \rightarrow \text{binary}$
1	Divide by 2, write quotient and remainder	$22 \div 2 = 11 \text{ R } 0$
2	Continue dividing quotient by 2	$11 \div 2 = 5 \text{ R } 1$
3	Continue until quotient is 0	$5 \div 2 = 2 \text{ R } 1$
4	Continue	$2 \div 2 = 1 \text{ R } 0$
5	Continue	$1 \div 2 = 0 \text{ R } 1$
6	Read remainders from bottom to top	10110_2

Visual:

```
2 | 22
2 | 11 R 0 ↑
2 | 5 R 1 |
2 | 2 R 1 |
2 | 1 R 0 |
  | 0 R 1 |
    101102
```

6. Converting Binary to Decimal

Method: Expand using powers of 2

Step	Action	Example: 1101_2
1	Identify place values	$2^3, 2^2, 2^1, 2^0$
2	Multiply each digit by its place value	$1 \times 8 + 1 \times 4 + 0 \times 2 + 1 \times 1$
3	Add the results	$8 + 4 + 0 + 1$
4	Write the decimal number	13_{10}

Visual:

$$\begin{aligned} 1101_2 &= 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 \\ &= 1 \times 8 + 1 \times 4 + 0 \times 2 + 1 \times 1 \\ &= 8 + 4 + 0 + 1 \\ &= 13_{10} \end{aligned}$$

7. Binary Addition Rules

Rule	Example
$0 + 0 = 0$	$0_2 + 0_2 = 0_2$
$0 + 1 = 1$	$0_2 + 1_2 = 1_2$
$1 + 0 = 1$	$1_2 + 0_2 = 1_2$
$1 + 1 = 10_2$ (carry 1)	$1_2 + 1_2 = 10_2$
$1 + 1 + 1 = 11_2$ (carry 1)	$1_2 + 1_2 + 1_2 = 11_2$

Quick Reference:

$$\begin{aligned} 1_2 + 0_2 &= 1_2 \\ 1_2 + 1_2 &= 10_2 \\ 1_2 + 1_2 + 1_2 &= 11_2 \end{aligned}$$

8. Binary Subtraction Rules

Rule	Example
$0 - 0 = 0$	$0_2 - 0_2 = 0_2$
$1 - 0 = 1$	$1_2 - 0_2 = 1_2$
$1 - 1 = 0$	$1_2 - 1_2 = 0_2$
$0 - 1 = \text{Borrow 1 from left}$	(Borrowing needed)

9. Key Relationships

Relationship	Example
$1 \text{ cm}^3 = 1 \text{ ml}$	Volume equivalence
$1 \text{ m}^3 = 1000 \text{ l}$	Volume equivalence
$10^2 = 100$	Power of 10
$2^3 = 8$	Power of 2
$2^0 = 1$	Any number to power 0 = 1

10. Check Yourself – Quick Questions

No.	Question	Answer
1	What are the digits used in binary?	0 and 1
2	What is the base of the binary number system?	2
3	What is the value of 2^5 ?	32
4	What is the value of 2^0 ?	1
5	What is the place value of the 4th position in binary?	8
6	What is 101_2 in decimal?	5
7	What is 10_2 in decimal?	2
8	What is 11_2 in decimal?	3
9	What is 100_2 in decimal?	4
10	What is 110_2 in decimal?	6
11	What is 111_2 in decimal?	7
12	What is 1000_2 in decimal?	8
13	What is $1_2 + 1_2$?	10_2
14	What is $10_2 + 1_2$?	11_2
15	What is $1_2 + 1_2 + 1_2$?	11_2

11. Quick Memory Aids

■ BINARY QUICK FACTS:

BASE = 2

DIGITS = 0, 1

PLACE VALUES = 1, 2, 4, 8, 16, 32, 64, 128...

■ POWERS OF 2 (MUST KNOW):

$$2^0 = 1$$

$$2^1 = 2$$

$$2^2 = 4$$

$$2^3 = 8$$

$$2^4 = 16$$

$$2^5 = 32$$

$$2^6 = 64$$

■ BINARY ADDITION:

$$1_2 + 0_2 = 1_2$$

$$1_2 + 1_2 = 10_2$$

$$1_2 + 1_2 + 1_2 = 11_2$$

■ WRITING NUMBERS:

Always write the base as a subscript:

101₂ (binary) 5₁₀ (decimal)

■ CONVERSION METHODS:

Decimal → Binary: Repeated division by 2

Binary → Decimal: Multiply by powers of 2

Revision Tip: Memorize powers of 2 up to 2^{10} (1024). Practice converting small numbers between decimal and binary. Use the repeated division method for decimal to binary — it's the easiest way to avoid mistakes!